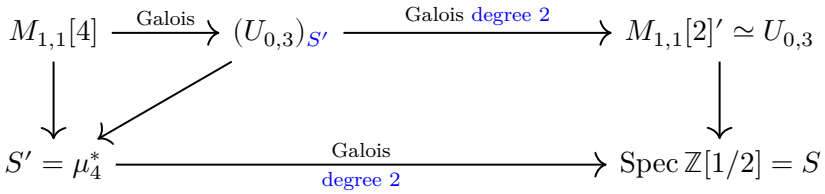
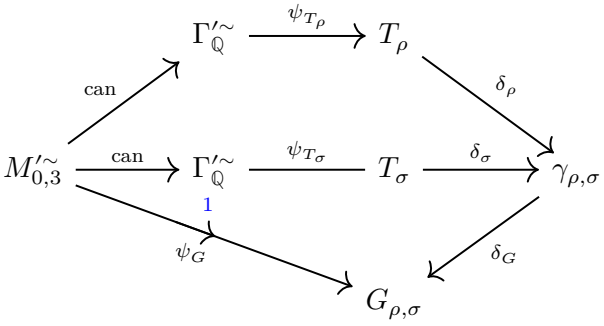
[scan 568]


$M_{1,1}[4]$
Galois
$(U_{0,3})_{S'}$
Galois degree 2
$M_{1,1}[2]' \simeq U_{0,3}$
$S' = \mu_4^*$
Galois
degree 2
$\operatorname{Spec} \mathbb{Z}[1/2] = S$

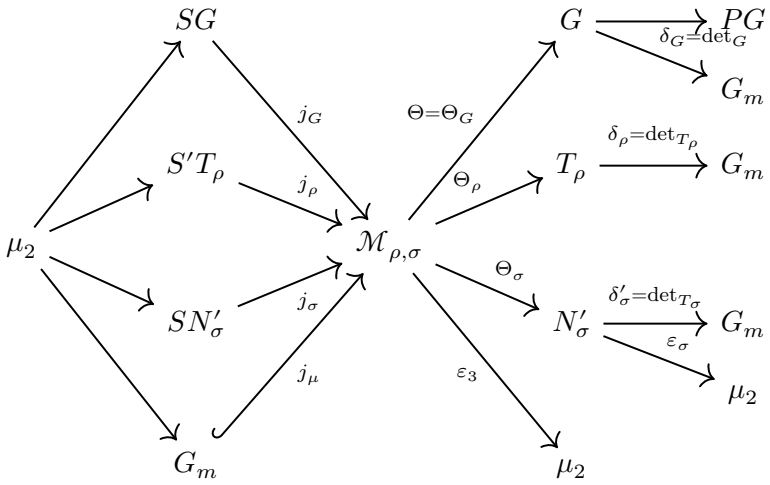
[scan 519]


$\Gamma'^{\sim}_{\mathbb{Q}}$
$\psi_{T_\rho}$
$T_\rho$
can
$\delta_\rho$
$M'^{\sim}_{0,3}$
can
$\Gamma'^{\sim}_{\mathbb{Q}}$
$\psi_{T_\sigma}$
$T_\sigma$
$\delta_\sigma$
$\gamma_{\rho,\sigma}$
1
$\psi_G$
$\delta_G$
$G_{\rho,\sigma}$

[scan 536]

$$
\begin{array}{ccccccccc}
1 & \to & L_0 & \to & B_{\rho,\sigma} & \xrightarrow{\delta_B} & \gamma_{\rho,\sigma} & \to & 1 \\
 & & \cup & & \downarrow & & \| & & \\
1 & \to & \mathcal{G} & \to & G_{\rho,\sigma} & \xrightarrow{\delta_G} & \gamma_{\rho,\sigma} & & \\
 & & & & & \searrow^{\chi_G} & \downarrow{\scriptstyle \chi_\gamma} & & \\
 & & & & [ & & \widehat{\mathbb{Z}}^* & & ]
\end{array}
$$

[scan 448]


$SG$
$S'T_\rho$
$SN'_\sigma$
$G_m$
$\mu_2$
$j_G$
$j_\rho$
$j_\sigma$
$j_\mu$
$\mathcal{M}_{\rho,\sigma}$
$\Theta{=}\Theta_G$
$\Theta_\rho$
$\Theta_\sigma$
$\varepsilon_3$
$G$
$PG$
$\delta_G{=}\det_G$
$G_m$
$T_\rho$
$\delta_\rho{=}\det_{T_\rho}$
$G_m$
$N'_\sigma$
$\delta'_\sigma{=}\det_{T_\sigma}$
$G_m$
$\varepsilon_\sigma$
$\mu_2$
$\mu_2$